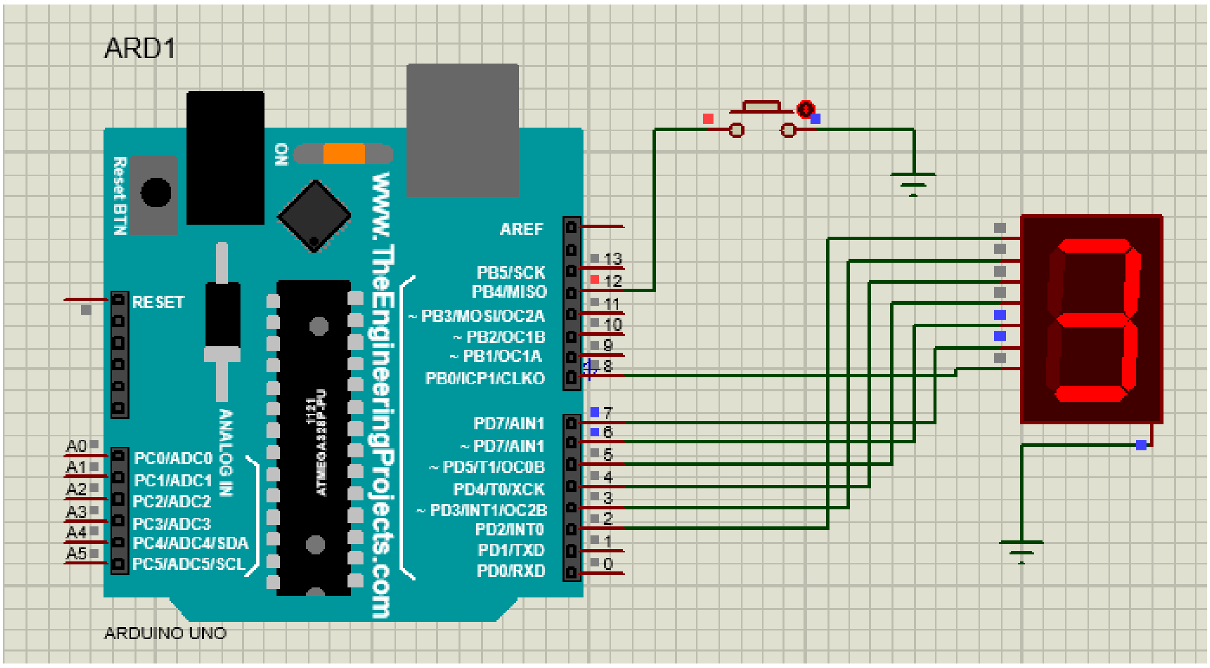
Exp No: 5: ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

CIRCUIT DIAGRAM:



PROGRAM:

```
// Explicitly initialize state variables to false

bool x0 = false, x1 = false, x2 = false, x3 = false, x4 = false;

bool x5 = false, x6 = false, x7 = false, x8 = false, x9 = false;

int delay_time = 200;

void setup() {

 pinMode(12, INPUT_PULLUP); // Pushbutton connected to Pin 12 and GND

 // Set Pins 2 to 8 as outputs for 7 segments (A, B, C, D, E, F, G)

 for (int pin = 2; pin <= 8; pin++) {

  pinMode(pin, OUTPUT);

 }

}

void loop() {

 int sensorVal = digitalRead(12);

 if (sensorVal == LOW && !x0 && !x1 && !x2 && !x3 && !x4 && !x5 && !x6 && !x7 && !x8 && !x9) {
```

```
  x0 = true;
  delay(200); // Simple debounce delay
}

while (x0) {
  zero();
  if (digitalRead(12) == LOW) { delay(200); x1 = true; x0 = false; }
}
while (x1) {
  one();
  if (digitalRead(12) == LOW) { delay(200); x2 = true; x1 = false; }
}
while (x2) {
  two();
  if (digitalRead(12) == LOW) { delay(200); x3 = true; x2 = false; }
}
while (x3) {
  three();
  if (digitalRead(12) == LOW) { delay(200); x4 = true; x3 = false; }
}
while (x4) {
  four();
  if (digitalRead(12) == LOW) { delay(200); x5 = true; x4 = false; }
}
while (x5) {
  five();
  if (digitalRead(12) == LOW) { delay(200); x6 = true; x5 = false; }
}
```

```
  while (x6) {
    six();
    if (digitalRead(12) == LOW) { delay(200); x7 = true; x6 = false; }
  }
  while (x7) {
    seven();
    if (digitalRead(12) == LOW) { delay(200); x8 = true; x7 = false; }
  }
  while (x8) {
    eight();
    if (digitalRead(12) == LOW) { delay(200); x9 = true; x8 = false; }
  }
  while (x9) {
    nine();
    if (digitalRead(12) == LOW) { delay(200); x0 = true; x9 = false; }
  }
}

// Fixed display functions mapped to Pins 2 through 8
void zero() {
  digitalWrite(2, HIGH); digitalWrite(3, HIGH); digitalWrite(4, HIGH);
  digitalWrite(5, HIGH); digitalWrite(6, HIGH); digitalWrite(7, HIGH);
  digitalWrite(8, LOW);
  delay(delay_time);
}

void one() {
  digitalWrite(2, LOW);  digitalWrite(3, HIGH); digitalWrite(4, HIGH);
```

```
  digitalWrite(5, LOW);  digitalWrite(6, LOW);  digitalWrite(7, LOW);
  digitalWrite(8, LOW);
  delay(delay_time);
}
void two() {
  digitalWrite(2, HIGH); digitalWrite(3, HIGH); digitalWrite(4, LOW);
  digitalWrite(5, HIGH); digitalWrite(6, HIGH); digitalWrite(7, LOW);
  digitalWrite(8, HIGH);
  delay(delay_time);
}
void three() {
  digitalWrite(2, HIGH); digitalWrite(3, HIGH); digitalWrite(4, HIGH);
  digitalWrite(5, HIGH); digitalWrite(6, LOW);  digitalWrite(7, LOW);
  digitalWrite(8, HIGH);
  delay(delay_time);
}
void four() {
  digitalWrite(2, LOW);  digitalWrite(3, HIGH); digitalWrite(4, HIGH);
  digitalWrite(5, LOW);  digitalWrite(6, LOW);  digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
  delay(delay_time);
}
void five() {
  digitalWrite(2, HIGH); digitalWrite(3, LOW);  digitalWrite(4, HIGH);
  digitalWrite(5, HIGH); digitalWrite(6, LOW);  digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
  delay(delay_time);
}
```

```
void six() {
  digitalWrite(2, HIGH); digitalWrite(3, LOW);  digitalWrite(4, HIGH);
  digitalWrite(5, HIGH); digitalWrite(6, HIGH); digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
  delay(delay_time);
}

void seven() {
  digitalWrite(2, HIGH); digitalWrite(3, HIGH); digitalWrite(4, HIGH);
  digitalWrite(5, LOW);  digitalWrite(6, LOW);  digitalWrite(7, LOW);
  digitalWrite(8, LOW);
  delay(delay_time);
}

void eight() {
  digitalWrite(2, HIGH); digitalWrite(3, HIGH); digitalWrite(4, HIGH);
  digitalWrite(5, HIGH); digitalWrite(6, HIGH); digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
  delay(delay_time);
}

void nine() {
  digitalWrite(2, HIGH); digitalWrite(3, HIGH); digitalWrite(4, HIGH);
  digitalWrite(5, HIGH); digitalWrite(6, LOW);  digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
  delay(delay_time);
}
```